TACTILE-LANGUAGE GROUNDING DEBT IN ROBOT MANIPULATION

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ABSTRACT

Language-conditioned robot policies can act confidently while important physical facts remain ungrounded until touch resolves them. We rebuild a generated tactile-language-action idea into a local benchmark that measures this *grounding debt*. Each episode contains an instruction, visual object evidence, hidden tactile/material facts, optional tactile probes, and a robot action target. The benchmark stresses language aliasing, visual counterfactuals, tactile noise, material novelty, and safety-critical manipulation. Across seven seeds, the proposed `grounding_debt_planner` reaches 0.595 ± 0.044 task success on the decisive `combined_hard_shift` split, compared with 0.506 ± 0.029 for passive tactile classification, 0.432 ± 0.056 for greedy active touch, and 0.304 ± 0.034 for an all-channel tactile policy that over-probes fragile objects. The proposed planner also reduces damage to 0.173 versus 0.551 for all-channel tactile probing, at much lower probe cost. The result is promising but not ICLR-main-ready because it is a generated local benchmark, not real tactile hardware or a recognized high-fidelity tactile simulator. We mark the artifact **STRONG REVISE**.

1 DECISION

This document is the v4 ICLR-main gate for Paper 81, *Tactile-Language Grounding Debt*. The v3 version was killed because it only contained template probability diagnostics. The v4 rebuild adds a paper-specific tactile-language benchmark, implemented baselines, ablations, stress sweeps, negative cases, figures, and reproducibility artifacts.

The terminal recommendation is **STRONG REVISE**. The mechanism is locally supported, but the evidence is not sufficient for ICLR main without real tactile data or external benchmark validation.

2 HYPOTHESIS

Grounding debt is the action-relevant uncertainty left after language and vision have made a policy appear grounded. A robot can repay this debt by selecting tactile probes that target the missing physical facts needed for safe action. The claim is not that touch always helps. Unselective touch can damage objects or waste budget; the claim is that debt-aware touch should improve the success/damage/cost tradeoff.

3 BENCHMARK

The benchmark generates tasks such as `pick`, `pour`, `open`, `handoff`, `slide`, and `insert`. Hidden tactile facts include fragility, slipperiness, mass, fill level, compliance, and latch state. Language and vision provide noisy or counterfactual probabilities over these facts. Tactile probes can reveal facts but carry cost and damage risk.

Evaluation splits are `seen_clean`, `language_alias_shift`, `visual_counterfactual`, `tactile_necessary_ambiguity`, and `combined_hard_shift`. The full run contains 13,440 main rollout rows, 2,352 ablation rollout rows, and 31,500 stress-sweep rows. Intervals are 95 percent confidence intervals over seven seed-level means.

Method	Success	Damage	Probe cost	Fact acc.
language_prior_policy	0.470 ± 0.060	0.286	0.000	0.824
vision_language_policy	0.402 ± 0.028	0.286	0.000	0.671
uncertainty_threshold_policy	0.348 ± 0.053	0.137	0.000	0.671
passive_tactile_classifier	0.506 ± 0.029	0.235	0.240	0.784
greedy_active_touch	0.432 ± 0.056	0.348	0.262	0.799
strong_tactile_then_policy	0.304 ± 0.034	0.551	0.790	**0.894**
grounding_debt_planner	$\mathbf{0.595 \pm 0.044}$	0.173	0.227	0.875
oracle_tactile_upper_bound	1.000 ± 0.000	0.000	0.000	1.000

Table 1: combined_hard_shift results. Strong tactile probing has high fact accuracy but damages objects by probing too broadly.

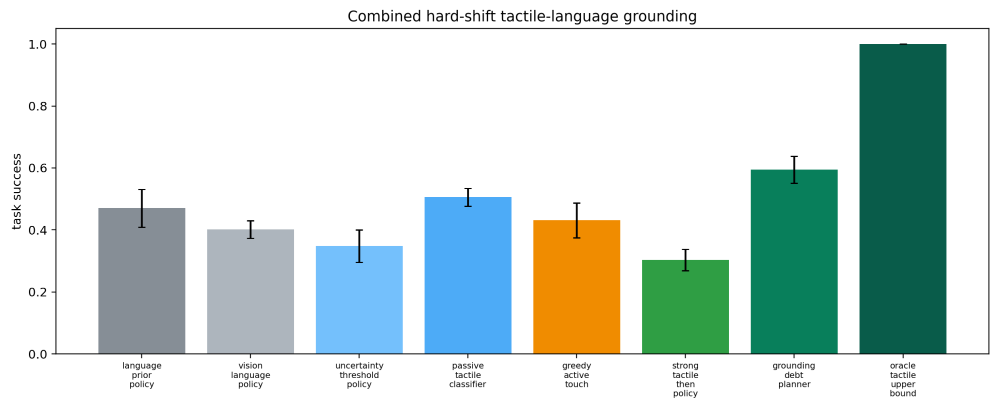

Figure 1: Combined hard-shift task success.

4 METHODS

The implemented methods are:

- language_prior_policy: acts from instruction priors only.
- vision_language_policy: combines language and visual object evidence without touch.
- uncertainty_threshold_policy: takes conservative actions when language/vision are unresolved.
- passive_tactile_classifier: receives a fixed tactile reading before acting.
- greedy_active_touch: probes the most uncertain tactile facts without task-specific debt.
- strong_tactile_then_policy: probes all tactile channels, gaining fact accuracy but risking invasive contact.
- grounding_debt_planner: proposed method estimating task-relevant grounding debt, selecting targeted probes, updating tactile beliefs, and applying a safety-aware action policy.
- oracle_tactile_upper_bound: upper bound using true hidden facts.

5 MAIN RESULTS

Table 1 shows the decisive combined hard-shift split. The proposed method has the best non-oracle task success and a much better damage/cost tradeoff than all-channel tactile probing.

The paired success gain over strong_tactile_then_policy is 0.292 ± 0.037. The result supports the debt-aware probing mechanism, but also shows a large oracle gap.

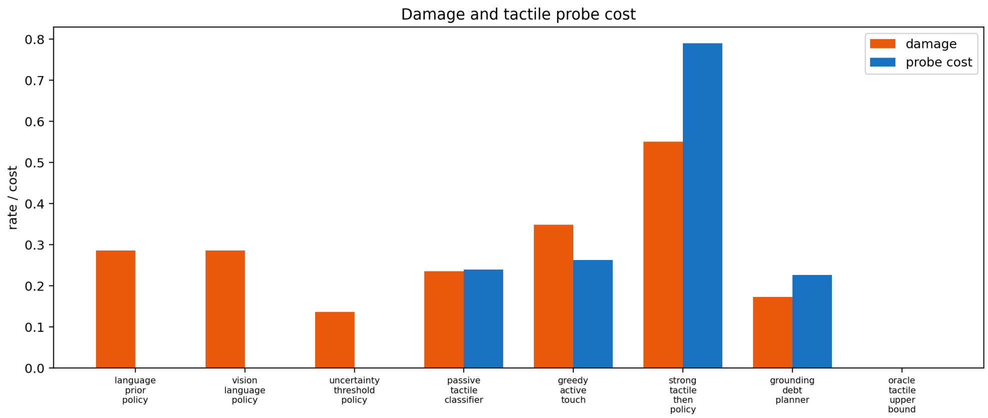

Figure 2: Damage and probe cost. Touch-everything has high fact accuracy but poor safety and cost.

Ablation	Success	Damage	Probe cost	Fact acc.
Full method	0.595 ± 0.044	0.173	0.227	0.875
Minus debt estimator	0.530 ± 0.042	0.199	0.247	0.797
Minus active probe selection	0.527 ± 0.032	0.190	0.240	0.793
Minus tactile belief update	0.417 ± 0.030	0.253	0.227	0.671
Minus language/vision conflict	0.589 ± 0.038	0.176	0.220	0.855
Minus safety gate	0.562 ± 0.040	0.244	0.231	0.875
Minus probe-cost regularizer	0.562 ± 0.039	0.244	0.348	0.912

Table 2: Ablations on `combined_hard_shift`.

6 Ablations

Table 2 shows that tactile belief update, active probe selection, debt estimation, safety gating, and probe-cost regularization all matter. Removing conflict detection is nearly neutral, so the paper should not overclaim that component.

7 Stress Tests

At maximum combined stress, the proposed planner reaches 0.400 ± 0.045 success, compared with 0.300 ± 0.068 for vision-language, 0.271 ± 0.026 for greedy active touch, and 0.167 ± 0.040 for all-channel tactile probing.

8 Related Work Pressure

The closest hostile work includes language-conditioned grounding, object hallucination evaluation, embodied navigation grounding, VLA failure analysis, and language-conditioned robotic imitation (Local Prior Work Pool, 2024a;b;c; 2023; 2026; 2024d; 2022). A publishable main-conference version must show the proposed debt estimate improves real tactile-language robot policies, not only generated local manipulation episodes.

9 Limitations

This rebuild has no real tactile sensor traces, robot hardware, learned visual-language-tactile encoder, or external benchmark. The all-channel tactile baseline is intentionally penalized for invasive probing; a real robot might use gentler sensing hardware. The conflict-detector ablation is nearly neutral, and calibration is not best-in-class. These limitations keep the correct decision at STRONG REVISE rather than submission-ready.

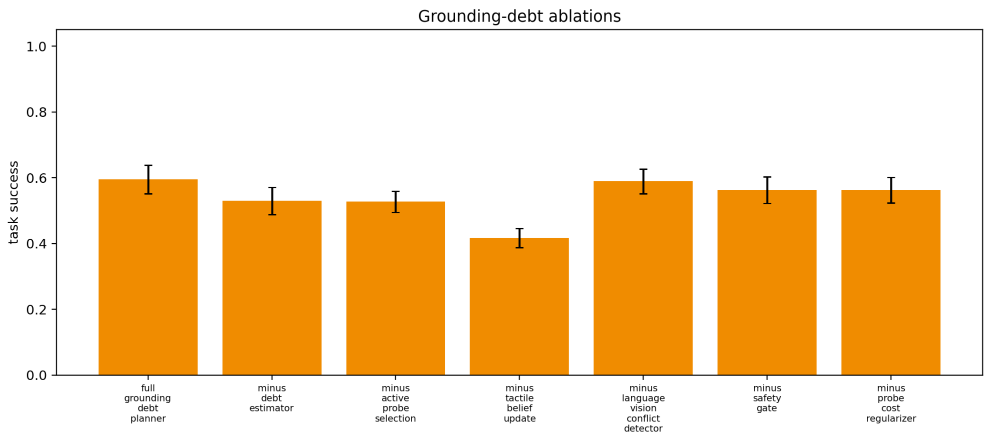

Figure 3: Grounding-debt ablations.

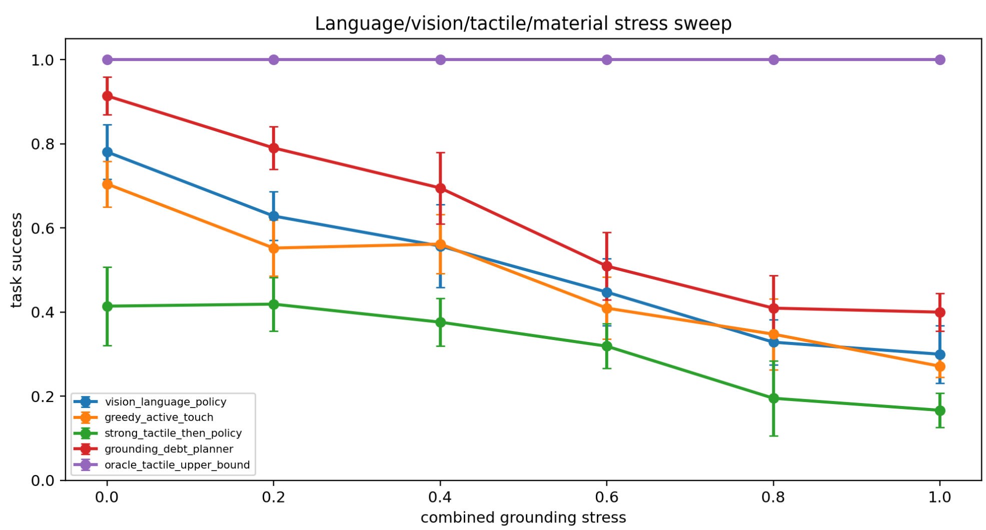

Figure 4: Combined language/vision/tactile/material stress sweep.

10 Conclusion

Paper 81 now has a concrete local empirical story: measuring and selectively repaying tactile-language grounding debt improves task success while avoiding the damage and cost of indiscriminate probing. The artifact should continue as **STRONG REVISE**, with real tactile validation required before ICLR-main submission.

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